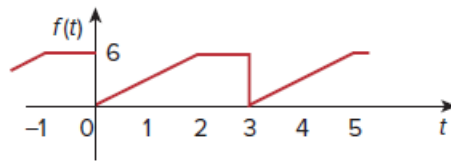


Problem

Find the effective value of $f(t)$ defined in Fig. 11.65.

Figure 11.65:



Step-by-step solution

Step 1 of 1

$$f_{rms}^2 = \frac{1}{T} \int_0^T f^2(t) dt$$

$$f_{rms}^2 = \frac{1}{3} \left[\int_0^3 (3t)^2 dt + \int_3^6 6^2 dt \right]$$

$$f_{rms}^2 = \frac{1}{3} \left[\frac{9t^3}{3} \Big|_0^3 + 36 \right]$$

$$f_{rms}^2 = 20$$

$$f_{rms} = \sqrt{20}$$

$$\therefore \boxed{f_{rms} = 4.472}$$